04/02/2026

Code A Phase 1

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 NCERT Booster Program for NEET-2026_NBTS-05A **Time : 180 Min.**

Topics Covered:
Physics: Electromagnetic Induction, Alternating Current, Electromagnetic Waves, Ray Optics & Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter
Chemistry: Chemical Kinetics, Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids
Botany: Molecular Basis of Inheritance, Microbes in Human Welfare, Organisms and Populations
Zoology: Reproductive Health, Evolution, Human Health and Disease

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. An LCR series circuit contains, $R = 50\ \Omega$, $L = 1$ mH and $C = 0.1\ \mu$F. The impedance of circuit will be minimum for a frequency of

   (1) $\frac{10^3}{2\pi}$ Hz

   (2) $\frac{10^4}{2\pi}$ Hz

   (3) $2\pi \times 10^5$ Hz

   (4) $2\pi \times 10^6$ Hz

2. A point source of light is placed at centre of curvature of hemispherical surface. The source emits a power of 36 W. The radius of curvature is 12 cm and inner surface is completely reflecting. The force on hemisphere due to light falling on it is $\alpha \times 10^{-8}$N, then the value of α is

   (1) 6

   (2) 4

   (3) 8

   (4) 24

3. The electric field of an electromagnetic wave in free space is given by $\vec{E} = 5\cos\left(10^7 t - kx\right)\hat{j}\frac{V}{m}$, where t is time in s and x is position in m. Choose the correct statement

   (1) The wavelength is 100 m

   (2) The value of k is 0.33 rad/m

   (3) The amplitude of electric field is $5\sqrt{2}$ V/m

   (4) The wave is propagating along $-x$ direction

4. Point A(0, 1 cm) and B(12 cm, 5 cm) are the coordinates of a real object and its image. If x-axis denotes the principal axis of the mirror, then this object-image pair is due to

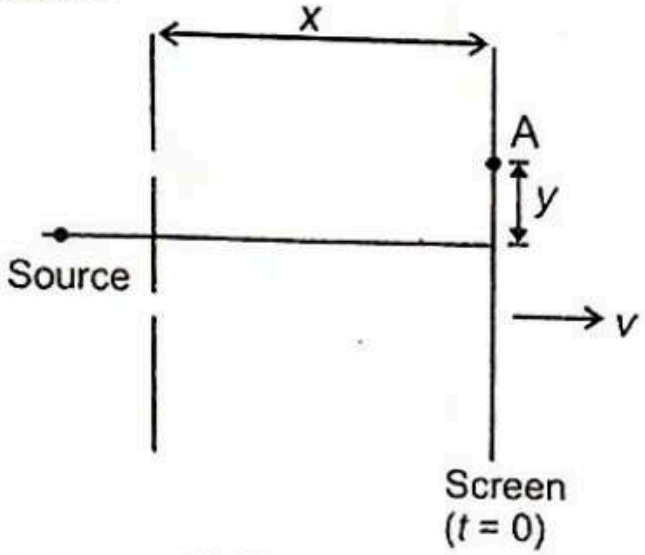
   (1) Concave mirror having its pole at (–2 cm, 0)

   (2) Concave mirror having its pole at (2 cm, 0)

   (3) Convex mirror having its pole at (2 cm, 0)

   (4) Convex mirror having its pole at (–3 cm, 0)

5. In the Young's arrangement, screen starts moving towards right with constant speed v. Initial distance between screen and plane of slits is x. At $t = 0$, 1st order maxima is lying at point A. After how much time first order minima lies at point A?



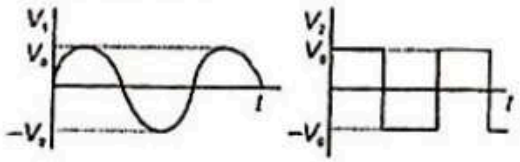
(1) $\frac{x}{2v}$

(2) $\frac{x}{v}$

(3) $\frac{x}{3v}$

(4) $\frac{2x}{3v}$

6. The sinusoidal potential difference V_1 shown in figure applied across a resistor R produces heat at rate W. What is the rate of heat dissipation when the square wave of potential difference V_2 as shown in figure is applied across the same resistor?



(1) $\frac{W}{2}$

(2) W

(3) $\sqrt{2}W$

(4) $2W$

7. A silver ball of radius 4.8 cm is suspended by a thread in the vacuum chamber. UV light of wavelength 200 nm is incident on the ball for some time during which a total energy of 1×10^{-7} joules falls on the surface. Assume that on an average one out of every ten thousand photons incident on the ball is able to eject an electron. The potential at centre of ball will be

(1) 1.2 V

(2) 6 V

(3) 3.1 V

(4) 0.3 V

8. If E_0 denotes peak emf, I_0 denotes peak current and Φ is phase difference between them, then average power dissipated in series LCR circuit will be

(1) $\frac{1}{2}E_0 I_0$

(2) $\frac{E_0 I_0}{2}\cos\phi$

(3) $\frac{E_0 I_0}{2}\sin\phi$

(4) $\frac{E_0 I_0}{\sqrt{2}}\cos\phi$

9. An EM wave propagates through a non-conducting medium with relative permittivity $\varepsilon_r = 9$ and relative permeability $\mu_r = 4$. The electric field amplitude is $E_0 = 90$ V/m. Find the amplitude of the magnetic field.

(1) 3×10^{-7} T

(2) 5.8×10^{-7} T

(3) 1×10^{-7} T

(4) 1.8×10^{-6} T

10. Consider the following statements out of which one is labelled as assertion and other as reason.
Assertion : If wavelength of light is doubled, energy and momentum of photons are reduced to half.
Reason : In vaccum, by increasing wavelength speed of photons will decrease.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

11. Match the columns and tick the correct option.

	Column I		Column II
a.	Simple microscope	(i)	Focal length of objective is more than eyepiece and final image is inverted
b.	Compound microscope	(ii)	Focal length of objective is less than eyepiece
c.	Astronomical telescope	(iii)	Have only one convex lens and produce virtual image
d.	Terrestrial telescope	(iv)	Produce magnified erect image of distant objects

(1) a(iii); b(i); c(ii); d(iv)

(2) a(iii); b(ii); c(i); d(iv)

(3) a(iv); b(ii); c(i); d(iii)

(4) a(iv); b(i); c(iii); d(ii)

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12. The number of turns in primary and secondary coils of a transformer is 50 and 200 respectively. If current in primary coil is 4 A, then current in secondary coil is

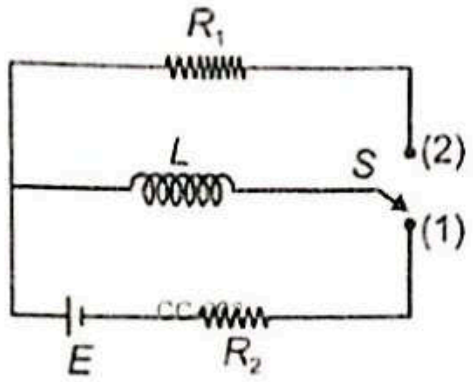
(1) 5 A

(2) 4 A

(3) 2 A

(4) 1 A

13. In the circuit as shown in figure, the switch is shifted from position 1 to position 2. The maximum amount of heat dissipated in R_1 is



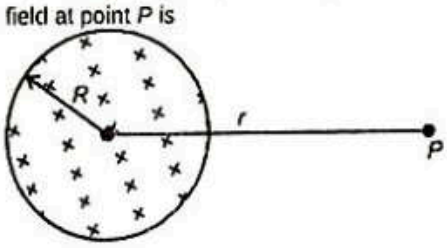
(1) $\frac{LE^2}{2R_1R_2}$

(2) $\frac{LE^2}{2R_2^2}$

(3) $\frac{LE^2}{2R_1^2}$

(4) Zero

14. The magnetic field in circular region of radius R varies with the rate of $\frac{dB}{dt}$ (tesla per second), then the induced electric field at point P is



(1) Zero

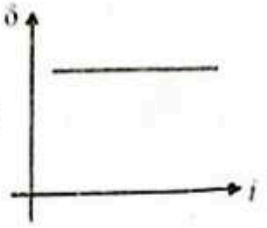
(2) $\frac{r}{R}\frac{dB}{dt}$

(3) $\frac{R^2}{r}\frac{dB}{dt}$

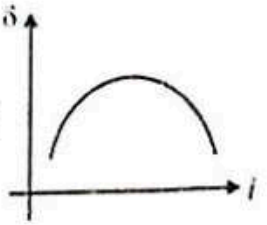
(4) $\frac{R^2}{2r}\frac{dB}{dt}$

15. The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is best represented by

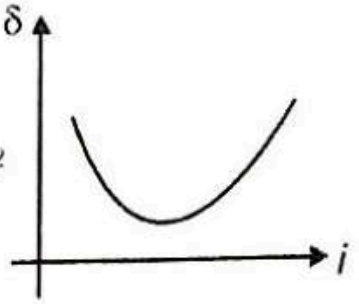
(1)



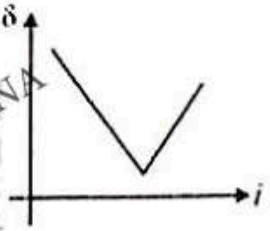
(2)



(3)



(4)



16. Two light waves having the same wavelength λ in vacuum are in phase initially. The first wave travels a path L_1 through a medium of refractive index n_1 while second wave travels a path of length L_2 through a medium of refractive index n_2. The two waves are then combined in vacuum to produce interference. The phase difference between the two waves is

(1) $\frac{2\pi}{\lambda}\left(L_2-L_1\right)$

(2) $\frac{2\pi}{\lambda}\left(n_1L_1-n_2L_2\right)$

(3) $\frac{2\pi}{\lambda}\left(n_2L_1-n_1L_2\right)$

(4) $\frac{2\pi}{\lambda}\left(\frac{L_1-L_2}{n_1-n_2}\right)$

17. If fringe width of YDSE is 0.4 mm, the distance between fifth bright and third dark band on same side of the central maximum, is

(1) 1 mm

(2) 4 mm

(3) 2 mm

(4) 3 mm

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18. In an ac series circuit, the instantaneous current in the circuit is maximum when the instantaneous voltage connected in the circuit is zero. The circuit may have a

(1) Pure inductor

(2) Pure capacitor

(3) Combination of inductor and capacitor

(4) All of these

19. Read the following statements.
Statement A: Self-inductance is called the inertia of electricity.
Statement B: Self-inductance and mutual inductance both have same S.I. unit.
Choose the correct option

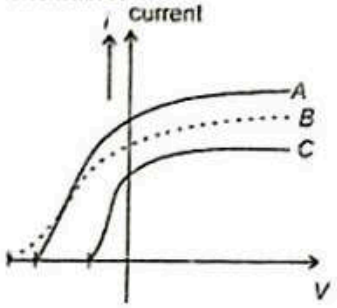
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct and statement B is incorrect

(4) Statement A is incorrect and statement B is correct

20. The figure shows the results of an experiment involving photoelectric effect. The graphs *A*, *B* and *C* relate to a light beam having different wavelengths. Select the correct alternative.



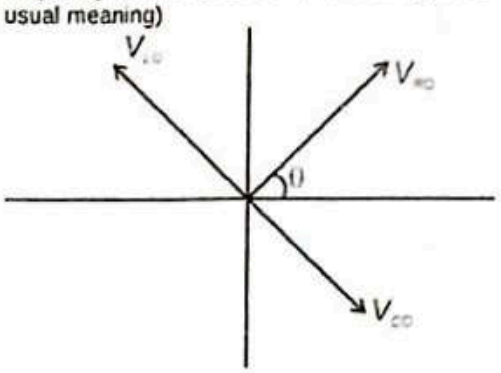
(1) Beam *B* has highest frequency

(2) Beam C has longest wavelength

(3) Beam A has highest rate of photoelectric emission

(4) All of these

21. In a series LCR circuit the phasors corresponding to voltage across resistance, capacitor and inductor at an instant are shown and have amplitudes of V_{RO} = 8 volt, V_{CO} = 3 volt and V_{LO} = 9 volt respectively. If V_{RO} phasor makes $\theta = 53°$ at $t = 0$, then correct variation of source voltage as a function of time (t) is (Assume angular frequency of source to be ω and all symbols have their usual meaning)



(1) 10 cos(ωt)

(2) 10 sin(ωt + 53°)

(3) 10 sin(ωt + 37°)

(4) 10 sinωt

22. For an EM wave propagating along *x*-axis in free space, the amplitude of oscillating electric field is 30 Vm^{-1}, then the amplitude of oscillating magnetic field will be

(1) 10^{-7} T

(2) 10^{-8} T

(3) 10^{-9} T

(4) 10^{-6} T

23. In the Young's double slit experiment using a monochromatic light of wavelength λ, the path difference corresponding to any point having half the peak intensity is (where *n* is an integer)

(1) $\left(2n+1\right)\frac{\lambda}{2}$

(2) $\left(2n+1\right)\frac{\lambda}{4}$

(3) $\left(2n+1\right)\frac{\lambda}{8}$

(4) $\left(2n+1\right)\frac{\lambda}{16}$

24. Particle *A* of mass $m_A = \frac{M}{2}$ moving along the x-axis with velocity v_0 collides elastically with another particle *B* at rest having mass $m_B = \frac{M}{3}$. If both the particles move along x-axis after collision, then the percentage change in de-Broglie wavelength of particle *A* is

(1) 400%

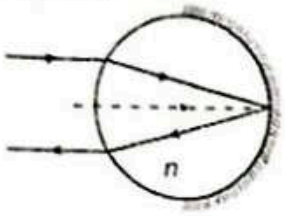
(2) 500%

(3) 300%

(4) 100%

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25. A transparent cylinder has its right half polished so as to act as a mirror. A paraxial light ray is incident from left, that is parallel to principal axis, exits parallel to the incident ray as shown. The refractive index n of the material of the cylinder is



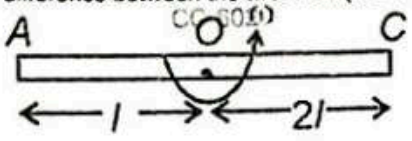
(1) 1.2

(2) 1.5

(3) 1.8

(4) 2.0

26. A conducting rod rotates with a constant angular velocity ω about the axis which passes through point O and perpendicular to its length. A uniform magnetic field B exists parallel to the axis of the rotation. Then potential difference between the two ends (A and C) of the rod is



(1) Zero

(2) $\frac{5}{2}B\omega l^2$

(3) $\frac{3}{2}B\omega l^2$

(4) $2B\omega l^2$

27. The average value of energy density associated with electric field of an EM wave is given by (where E_0 denotes peak value of electric field)

(1) $\frac{1}{2}\varepsilon_0 E_0^2$

(2) $\frac{E_0^2}{2\varepsilon_0}$

(3) $\frac{1}{4}\varepsilon_0 E_0^2$

(4) $\varepsilon_0 E_0^2$

28. A ray of light is incident on a plane mirror at an angle of 60°. The angle of deviation produced by mirror is

(1) 120°

(2) 30°

(3) 60°

(4) 90°

29. Match List-I with List-II

	List-I		List-II
a.	Frequency of incident radiation increases	(i)	Stopping potential decreases
b.	Number of incident photons increases	(ii)	Stopping potential increases
c.	Intensity of incident radiation decreases	(iii)	Saturation current increases
d.	Maximum K.E. of photoelectrons decreases	(iv)	Number of photoelectrons emitted per second decreases

Choose the correct answer from the options given below:

(1) a(iii), b(ii), c(i), d(iv)

(2) a(i), b(iv), c(iii), d(ii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iv), b(i), c(ii), d(iii)

30. In a compound microscope, focal length of objective and eyepiece are 2 cm and 5 cm respectively. An object is placed at a distance 2.2 cm from objective. If final image is formed at least distance of distinct vision (25 cm), then magnifying power of microscope and approximate distance between objective and eyepiece will be respectively

(1) 60, 18 cm

(2) 60, 26 cm

(3) 50, 20 cm

(4) 50, 26 cm

31. A current passing through a coil of self inductance 3 mH changes at the rate of 20 mA/s, then emf induced in coil is

(1) 10 μV

(2) 60 μV

(3) 10 mV

(4) 40 mV

32. Consider the following statements about electromagnetic waves:
a. They are longitudinal in nature.
b. They require a medium for propagation.
c. Electric and magnetic field components oscillate perpendicular to each other.
d. They travel with same speed in vacuum irrespective of their wavelengths.
The correct statement(s) is/are

(1) Only c

(2) Both c and d

(3) Only d

(4) Both b and c

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33. For Young's double slit experiment, two statements are given below:

Statement I: If slit separation increases, then angular width of central bright fringes remains constant.

Statement II: If the monochromatic source is replaced by another monochromatic source of smaller wavelength, the angular width of central bright fringes decreases.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are false

(2) Both statement I and statement II are true

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

34. Consider the following statement regarding Lenz's law of electromagnetic induction

(A) : The emf is induced in direction supporting the change in magnetic flux.

(B): The relative motion between magnet and coil produces change in magnetic flux

Based on above information pick correct option

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

35. The magnetic flux linked with coil varies as, $\Phi = 5t^2 + 2t$ where t denotes time in second and Φ is in weber. At $t = 1$ s, the value of induced emf is

(1) 14 V

(2) 10 V

(3) 12 V

(4) Zero

36. The photons in radiowave of wavelength 3×10^4 cm has energy equal to

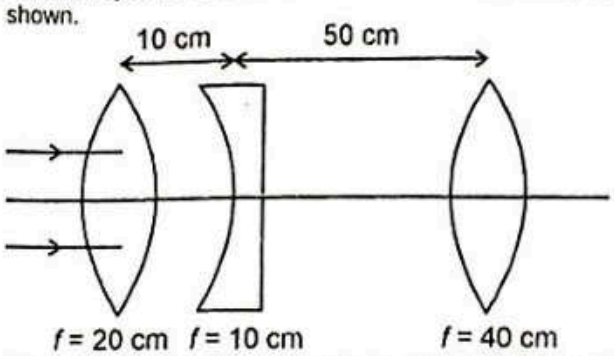
(1) 2.2×10^{-35} J

(2) 6.62×10^{-28} J

(3) 6.62×10^{-10} J

(4) 19.86×10^{-24} J

37. Parallel rays are focused on a combination of lenses as shown.



After refraction from final convex lens, the final image will be formed at a distance (from convex lens of focal length 40 cm):

(1) 10 cm

(2) 40 cm

(3) 80 cm

(4) Infinity

38. A series LCR circuit having resistance R, capacitance C and inductance L has a voltage source of angular frequency ω and input voltage V_{in}. If output voltage (V_{out}) is taken as voltage across resistor and inductor combined, then the ratio $\frac{V_{out}}{V_{in}}$ will be equal to

(1) $\sqrt{\frac{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}{R^2+(\omega L)^2}}$

(2) $\sqrt{\frac{R^2+\omega^2L^2}{R^2-\left(\omega L-\frac{1}{\omega C}\right)^2}}$

(3) $\sqrt{\frac{R^2+\left(\omega L+\frac{1}{\omega C}\right)^2}{R^2+(\omega L)^2}}$

(4) $\sqrt{\frac{R^2+\omega^2L^2}{R^2\left(\omega L-\frac{1}{\omega C}\right)^2}}$

39. The net magnetic flux through any closed surface, kept in a magnetic field is (Symbols have their usual meaning)

(1) Zero

(2) $\frac{\mu_0}{4\pi}$

(3) $\frac{\mu_0}{2\pi}$

(4) $\frac{4\mu_0}{\pi}$

40. A fish in water (refractive index = μ) looks at a bird vertically above in the air. If y is the height of bird from the surface of water and x is depth of fish from the surface, then distance of bird as estimated by fish is

(1) $x + y - \frac{y}{\mu}$

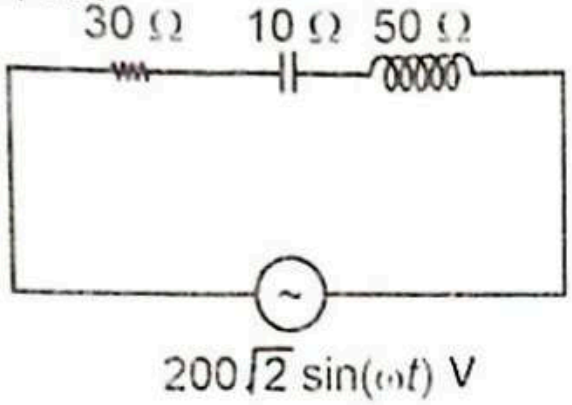
(2) $x + y + \frac{y}{\mu}$

(3) $x + \mu y$

(4) $x - \mu y$

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41. In the circuit shown below, the inductor and capacitor are ideal, then the incorrect statement among the given options is



(1) The RMS value of current in the circuit is 4 A

(2) The phase difference between current and source voltage is $\cos^{-1}\left(\frac{4}{5}\right)$

(3) Power consumed in the circuit is 480 W

(4) RMS value of the source voltage is 200 V

42. The Maxwell equation $\oint \vec{B}.\vec{dl} = \mu_0\left(i_c + \varepsilon_0\frac{d\Phi_E}{dt}\right)$ is a statement of (All symbols have their usual meaning)

(1) Faraday's law

(2) Gauss law of magnetism

(3) Modified Ampere's law

(4) Gauss law of electricity

43. Refraction of plane wavefront as per Huygen's principle are as shown in the figures. Figure (i) indicates refraction from air to glass and figure (ii) indicates refraction from air to water. In terms of lengths indicated in figures, the refractive index of glass w.r.t. water is

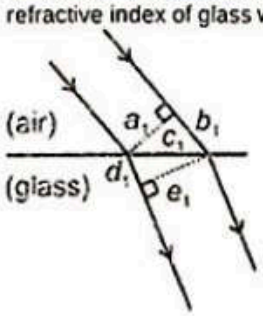


Figure (i)

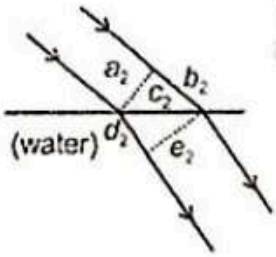


Figure (ii)

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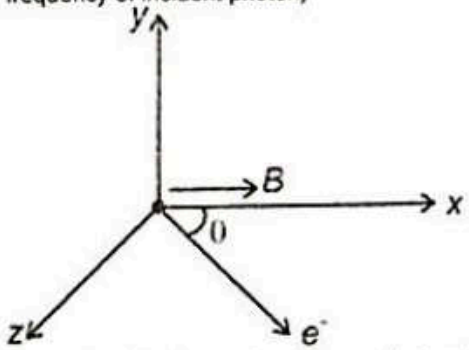
(1) $\frac{b_1}{b_2}$

(2) $\frac{d_1}{d_2}$

(3) $\frac{b_1d_1}{b_2d_2}$

(4) $\frac{b_1d_2}{b_2d_1}$

44. Light having photon energy (hf) is incident on a metallic plate having work function Φ_0 to eject the electrons. The most energetic electrons are then allowed to enter in a region of uniform magnetic field B. The electrons entered in x-z plane making an angle θ with x-axis and magnetic field $\vec{B} = B_0\hat{i}$ along x-axis as shown (where f denotes frequency of incident photon)



The pitch of helix described by an electron is found to be P. Assuming the mass of electron as m and charge as e, the correct relation between P and B_0 is

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(1) $PeB_0 = 2\pi\cos\theta\sqrt{2(hf-\phi_0)m}$

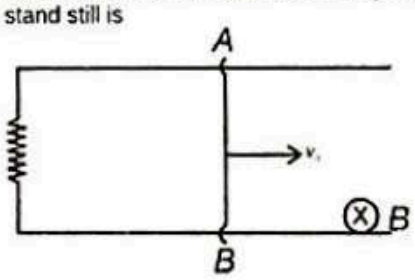
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(2) $PeB_0 = 2\pi\cos\theta\sqrt{\frac{2(hf-\phi_0)}{m}}$

(3) $PeB_0 = 2\pi\sqrt{2(hf-\phi_0)m}$

(4) $PeB_0 = 2\pi\sin\theta\sqrt{2(hf-\phi_0)m}$

45. A conducting rod AB of mass m slides without friction over two long conducting rails separated by distance l. At the left end the rails are interconnected by a resistance R. The system is located in uniform magnetic field perpendicular to plane of loop. At the moment, $t = 0$ the rod AB starts moving to right with initial velocity v_0. Neglecting the resistance of the rails and rod AB as well as as self inductance, the distance covered by rod until it comes to stand still is



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(1) $\frac{v_0mR}{B^2l}$

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(2) $\frac{v_0mR}{B^2l^2}$

(3) $\frac{2v_0mR}{B^2l^2}$

(4) $\frac{v_0mR}{2B^2l^2}$

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46. For a first order reaction, if $t_{1/2}$ is 40 s then $t_{87.5\%}$ will be

(1) 60 s

(2) 80 s

(3) 120 s

(4) 160 s

47. The half-life of decomposition of N_2O by first order kinetics is 30 min. If the initial concentration of reactant is 100 g, then calculate the approximate time required for 90% completion of the reaction.

(1) 80 min

(2) 100 min

(3) 90 min

(4) 115 min

48. The variation of rate constant with temperature T is given by Arrhenius equation. The rate constant at $T \to \infty$ will be

(1) A

(2) $\frac{E_a}{R}$

(3) $\frac{1}{A}$

(4) $-\frac{E_a}{R}$

49. In the following reaction, $C_6H_5Br \xrightarrow[ii)\ C_2H_5OH]{i)\ Mg,\ ether} A$, the major product (A) is

(1) C_6H_6

(2) $C_6H_5-C_2H_5$

(3) C_6H_5-OH

(4) $C_6H_5OC_2H_5$

50. Given below are the two statements.
Statement I : Propan-2-ol is an achiral molecule.
Statement II : Butan-2-ol is a chiral molecule.
In light of the above statements, choose the correct answer

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

51. Consider the following name reactions
(a) Fittig reaction
(b) Wurtz reaction
(c) Finkelstein reaction
The reactions which results in formation of C –C bond are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

52. Given below are the two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):?
Assertion (A) : Reaction of propene with HBr in presence of benzoyl peroxide gives 2-bromopropane as major product
Reason (R) : Reaction of propene with HBr in presence of benzoyl peroxide follows free-radical mechanism.
In light of above statements, choose the correct answer

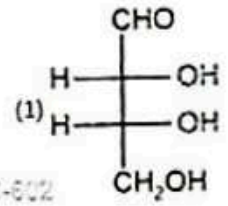
(1) (A) is false but (R) is true

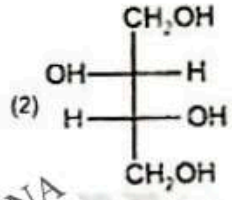
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

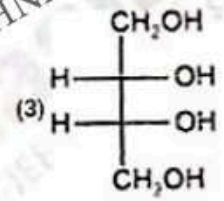
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

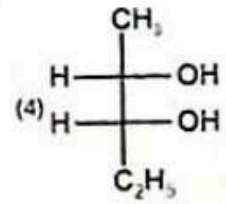
(4) (A) is true but (R) is false

53. Which of the following is a meso compound?

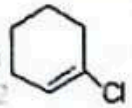
(1) 

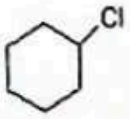
(2) 

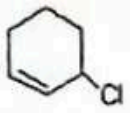
(3) 

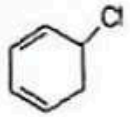
(4) 

54. Elimination reaction by alc. KOH will be fastest in

(1) 

(2) 

(3) 

(4) 

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55. The pair of compounds which cannot be distinguished by I_2/NaOH is

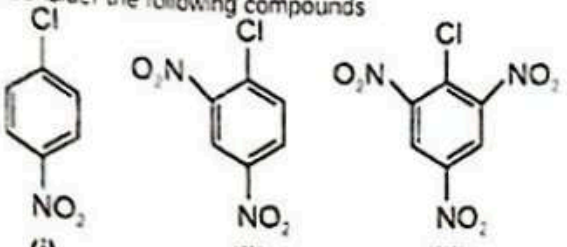
(1) Acetone and methanol

(2) Ethanol and benzaldehyde

(3) Acetophenone and benzophenone

(4) Acetaldehyde and isopropyl alcohol

56. Consider the following compounds



(i) (ii) (iii)

The correct order of reactivity for nucleophilic substitution reaction is

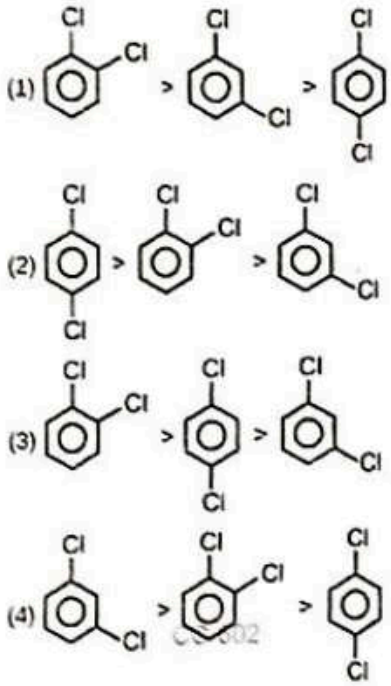
(1) (i) > (ii) > (iii)

(2) (iii) > (ii) > (i)

(3) (ii) > (i) > (iii)

(4) (iii) > (i) > (ii)

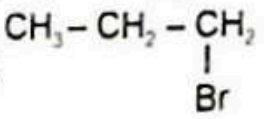
57. Arrange the following species in their decreasing order of melting points



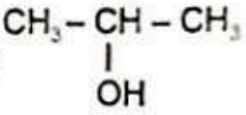
58. Consider the following reaction.

Propan-1-ol $\xrightarrow{PBr_3}$ X $\xrightarrow[\Delta]{alc.\ KOH}$ Y (Major) $\xrightarrow{}$ Z (Major) final product Z is

(1) $CH_3 - CH_2 - CH_2 - HSO_4$

(2) 

(3) $CH_3 - CH_2 - OCH_3$

(4) 

59. In the following question, a statement of assertion (A) is followed by a statement of reason (R)
Assertion : Phenol is more acidic than ethyl alcohol.
Reason : Phenoxide ion is resonance stabilised.
In the light of above statement choose the correct answer

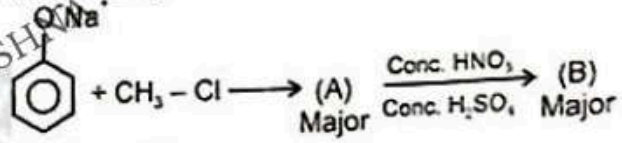
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

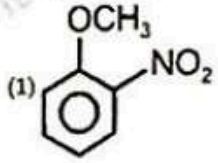
(3) Assertion is true statement but Reason is false

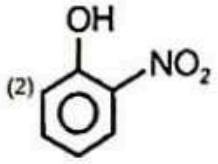
(4) Both Assertion and Reason are false statements

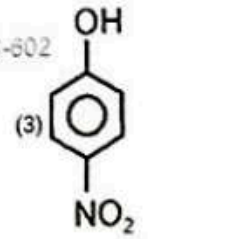
60. In the following reaction sequence

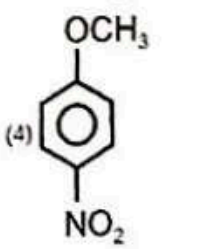


The product (B) is

(1) 

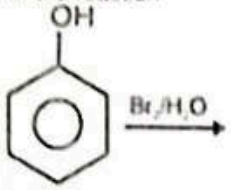
(2) 

(3) 

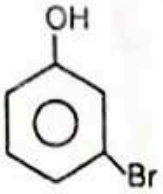
(4) 

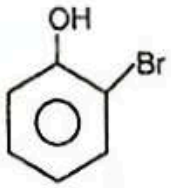
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61. In the reaction

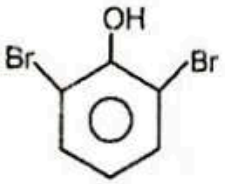


The major product formed is

(1) 

(2) 

(3) 

(4) 

62. Which of the following enzymes, converts glucose to alcohol?

(1) Zymase

(2) Maltose

(3) Diastase

(4) Invertase

63. The compound which will give immediate turbidity on reaction with Lucas reagent is

(1) 3-Methylpentan-2-ol

(2) 3,3-Dimethylbutan-1-ol

(3) 2-Methylpentan-2-ol

(4) 4-Methylpentan-2-ol

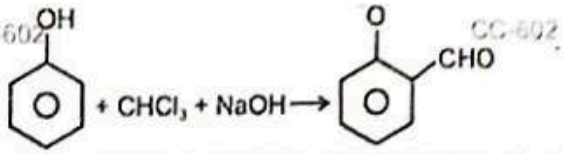
64. When anisole is treated with one equivalent HI, the products formed are

(1)  and CH_3–OH

(2) CH_3–I and 

(3) CH_3–I and 

(4) C_2H_5–OH and 

65. 

Above reaction proceeds via formation of which electrophile?

(1) $\ddot{C}H_2$

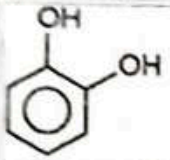
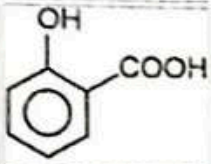
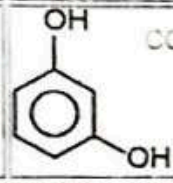
(2) $\dot{C}l$

(3) $:CCl_2$

(4) $\dot{C}Cl_3$

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66. Match List-I with List-II

	List I (Structure)		List-II (Common name)
a.		(i)	Resorcinol
b.		(ii)	Quinol
c.		(iii)	Catechol
d.		(iv)	Salicylic acid

Choose the correct answer from the options given below

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iii), b(iv), c(i), d(ii)

67. Time of completion of zero order reaction with initial concentration 'a M' and rate constant k mol L^{-1} s^{-1} is

(1) $\frac{a}{k}$

(2) $\frac{2a}{k}$

(3) $\frac{a}{2k}$

(4) ak

68. Unit of rate of reaction for second order reaction is

(1) s^{-1}

(2) mol^{-1} L s^{-1}

(3) mol L^{-1} s^{-1}

(4) mol^{-1} L s

69. If rate of reaction gets doubled on increase of every 10°C rise of temperature then by what factor rate of reaction will change if temperature is increased from 30°C to 60°C

(1) Increased by 6 times

(2) Decreased by 6 times

(3) Increased by 8 times

(4) Decreased by 8 times

70. Given below are two statements

Statement (I) : Hydrolysis of cane sugar in acidic medium is a zero order reaction.

Statement (II) : Rate of reaction is temperature independent for zero order reaction.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

71. For a chemical reaction rate of reaction doubles on increasing temperature from 300 K to 310 K, then activation energy for the reaction will be

(1) 18.4 kJ

(2) 106.8 kJ

(3) 24.44 kJ

(4) 53.42 kJ

72. Mechanism of a hypothetical reaction, CC-602

$2A_3 \rightarrow 3A_2$

is given below

(I) $A_3 \rightleftharpoons A_2 + A$ (Fast)

(II) $A_3 + A \rightarrow 2A_2$ (Slow)

Rate law for reaction will be

(1) $r = k[A_3]^2 [A_2]^{-1}$

(2) $r = k[A_3][A_2]$

(3) $r = k[A_3]^2 [A_2]$

(4) $r = k[A_3]^2$

73. $C_2H_5Cl \xrightarrow{KCN} A \xrightarrow[\text{(Major)}]{H^+/H_2O} B \text{ (Major)} \xrightarrow{Cl_2/Red\ P} C\text{(Major)}$

Product 'C' formed in above reaction sequence is

(1) C_3H_7Cl

(2) C_2H_5COCl

(3) $CH_3CHClCOOH$

(4) $CH_2(Cl)CH_2COOH$

74. $C_6H_5CHO + C_6H_5COCH_3 \xrightarrow[\Delta]{NaOH} \underset{\text{(Major)}}{P}$

Product 'P' formed in the reaction is

(1) C_6H_5COONa

(2) $C_6H_5CH_2OH$

(3) $C_6H_5CHCHCOC_6H_5$

(4) $C_6H_5CH(OH)CH_2COC_6H_5$

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75. Correct decreasing order of pK_a for given acids is

(a) CH_3COOH

(b) $HCOOH$

(c) C_6H_5COOH

(1) (a) > (b) > (c)

(2) (b) > (c) > (a)

(3) (a) > (c) > (b)

(4) (c) > (b) > (a)

76. Given below are two statements one is assertion (A) other is reason (R)

Assertion (A) : CF_3COOH is more acidic than CCl_3COOH

Reason (R) : 'F' has higher electron withdrawing effect than 'Cl'

In the light of above statements, choose the most appropriate answer from the options given below

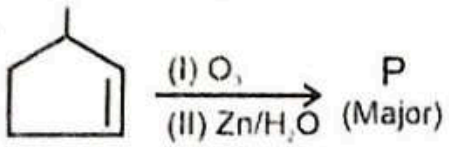
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

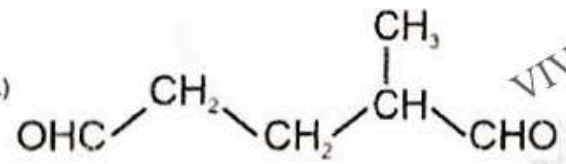
(4) (A) is false but (R) is true

77.

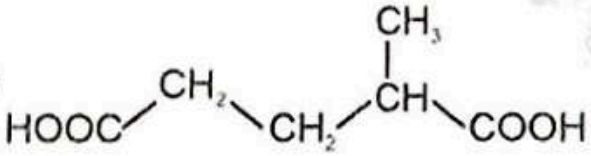


Product P formed in given reaction

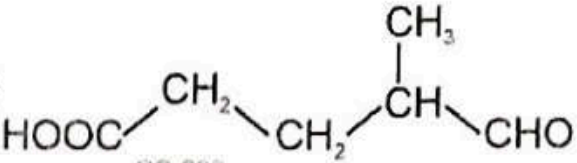
(1)



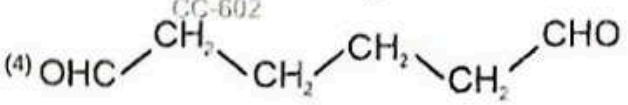
(2)



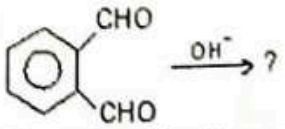
(3)



(4)

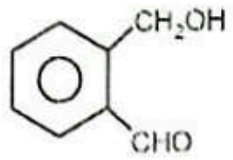


78.

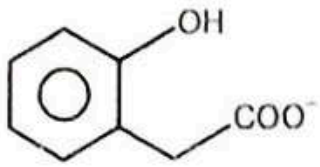


The product of the above reaction is

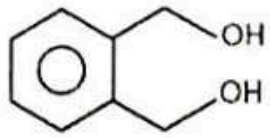
(1)



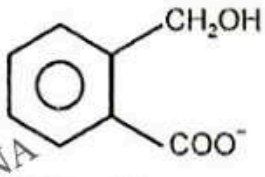
(2)



(3)



(4)



79. Reaction of acetone with Grignard reagent followed by hydrolysis gives

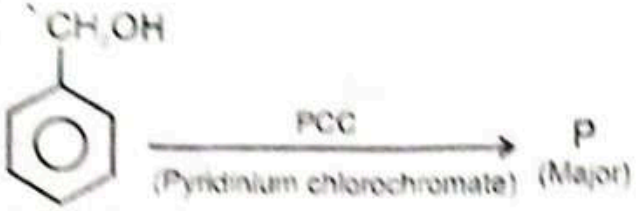
(1) Tertiary alcohol

(2) Carboxylic acid

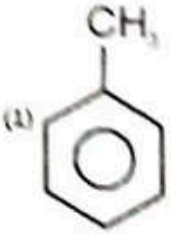
(3) Secondary alcohol

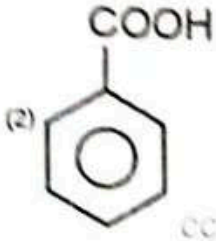
(4) Alkane

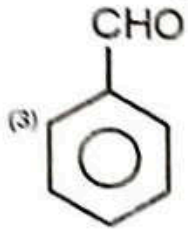
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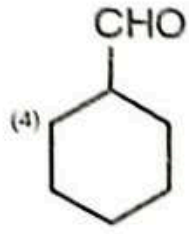
80. 

Product 'P' will be

(1) 

(2) 

(3) 

(4) 

81. Optically active monocarboxylic acid among the following is

(1) Succinic acid

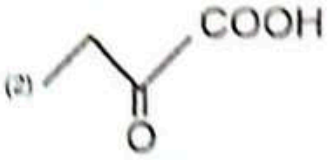
(2) Adipic acid

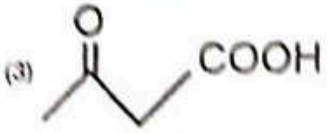
(3) Lactic acid

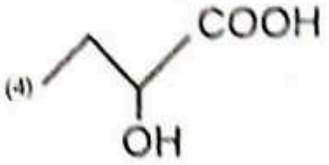
(4) Malonic acid

82. Which among the following will get decarboxylated most easily on heating with soda lime

(1) 

(2) 

(3) 

(4) 

83. Consider the following statements

Statement I : Reaction of benzoic acid with $NaHCO_3$ gives CO_2 gas.

Statement II : Benzoic acid is more acidic than picric acid.

In the light of above two statements choose the correct option

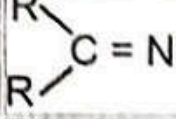
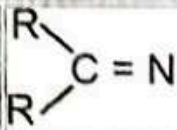
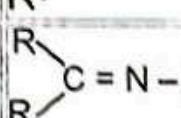
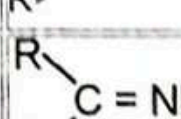
(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

84. Match list I with list II for N-substituted derivatives of aldehyde and ketones

	List-I Carbonyl derivative		List-II Product name
a.		(i)	Imine
b.		(ii)	Phenyl hydrazone
c.		(iii)	Schiff base
d.		(iv)	Hydrazone

Correct match is

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iv), b(i), c(ii), d(iii)

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85. Acrolein (Prop-2-en-1-al) is aldol condensation product of

(1) $CH_3\overset{O}{\overset{\|}{C}}CH_3$ and HCHO

(2) CH_3CHO and C_6H_5CHO

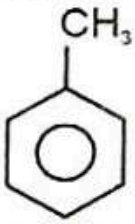
(3) $C_6H_5-\overset{O}{\overset{\|}{C}}-CH_3$ and HCHO

(4) CH_3CHO and HCHO

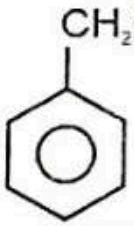
86. $O=C-CH_3$ 

$\xrightarrow[HCl]{Zn\text{-}Hg}$ P (Major)

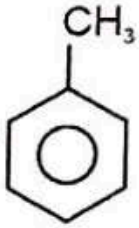
product 'P' and name of the reaction respectively will be

(1) 

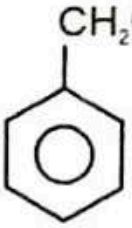
and Wolff-Kishner reduction

(2) 

and Wolff-Kishner reduction

(3) 

and Clemmensen reduction

(4) 

and Clemmensen reduction

87. Consider the following compounds

$CHCl_2-\overset{O}{\overset{\|}{C}}-OCH_3$ (I), $CH_3-\overset{O}{\overset{\|}{C}}-OCH_3$ (II), $CH_2Cl-\overset{O}{\overset{\|}{C}}-OCH_3$ (III)

Correct order of the hydrolysis of esters in alkaline medium is

(1) (I) > (II) > (III)

(2) (I) > (III) > (II)

(3) (II) > (III) > (I)

(4) (III) > (II) > (I)

88. Match reactions in list-I with their name in list-II

	List-I		List-II
a.	$RCN \xrightarrow[(II)\ H_2O]{(I)\ SnCl_2/HCl}$	(i)	Etard reaction
b.	 $\xrightarrow[(II)\ H_3O^+]{(I)\ CrO_2Cl_2/CS_2}$	(ii)	Cannizzaro reaction
c.	$HCHO \xrightarrow{Conc.KOH}$	(iii)	Hell-Volhard-Zelinsky reaction
d.	$RCH_2COOH \xrightarrow[(ii)\ H_2O]{(i)\ Red\ Phosphorous/Br_2}$	(iv)	Stephen reaction

Correct Match is

(1) a(iv), b(i), c(ii), d(iii)

(2) a(i), b(iv), c(ii), d(iii)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(i), b(iv), c(iii), d(ii)

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89. Given below are two statements.

Statement I : Carboxylic acids are higher boiling liquids than aldehydes, ketones and even alcohols of comparable molecular masses.

Statement II : Simple aliphatic carboxylic acids having upto four carbon atoms are miscible in water

In the light of the above statements, choose the most appropriate answer from the options given below

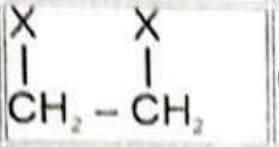
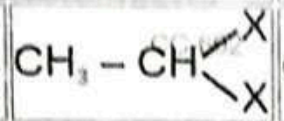
(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

90. Match halogen derivatives in list-I with their classification in list-II

	List-I		List-II
a.		(i)	Allyl halide
b.		(ii)	Vinyl halide
c.	$CH_2 = CH - CH_2Cl$	(iii)	Vicinal dihalide
d.	$CH_2 = CH\,Cl$	(iv)	Geminal dihalide

Correct match is

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(iv), c(ii), d(i)

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91. During the formation of bread, it becomes porous due to the release of CO_2 by the action of

(1) *Lactobacillus*

(2) *Monascus purpureus*

(3) *Saccharomyces cerevisiae*

(4) *Acetobacter aceti*

92. Read the following features.

(a) Form nodules in legume plants.

(b) Enhance nitrogen availability for the plants

(c) Show host-specific symbiosis

(d) Used as biofertilisers

How many of the above features is/are **true** for both *Rhizobium* and *Azospirillum*?

(1) Three

(2) One

(3) Four

(4) Two

93. Which of the following is further digested in the anaerobic sludge digester during the sewage treatment process?

(1) Effluent from primary treatment

(2) Sedimented bacterial flocs

(3) Primary sludge

(4) Effluent from biological treatment

94. Which of the following features are **true** w.r.t. *Methanobacterium*?
(a) Plays an important role in the nourishment of cattle.
(b) Grows aerobically to break down cellulose rich food.
(c) Responsible for the production of a gaseous fuel.
(d) Involved in processes, carried out in an STP
(e) These are eubacterium growing in marshes.
Choose the **correct** option.

(1) (a), (c) and (d) only

(2) (a), (c) and (e) only

(3) (b) and (e) only

(4) (b), (c) and (d) only

95. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A) : Activated sludge formed during the sewage treatment is rich in aerobic microbes.
Reason (R) : Aerobic microbes utilize dissolved oxygen for the oxidation of organic matter in the sewage

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Only (A) is true

(4) Only (R) is true

96. A common biocontrol agent for the control of plant diseases is

(1) *Nucleopolyhedrovirus*

(2) *Trichoderma*

(3) *Bacillus thuringiensis*

(4) *Glomus*

97. Consider the following statements and select the **correct** option
Statement (A) : Yeast is used in the commercial production of blood cholesterol lowering agents.
Statement (B) : *Staphylococcus* does not show growth in the vicinity of *Penicillium*.

(1) Both the given statements are correct

(2) Both the given statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

98. What will be the most appropriate consequence if the constant agitation and pumping of air during the secondary treatment is stopped?

(1) Level of BOD will drop down

(2) Organic matter will accumulate

(3) Least production of methane will occur

(4) Vigorous floc formation will take place

99. Choose the **incorrect** match from the following

(1) Aspergillus	Acetic acid
(2) Clostridium	Butyric acid
(3) Saccharomyces	Ethanol
(4) Trichoderma	Cyclosporin A

(1) (1)

(2) (2)

(3) (3)

(4) (4)

100. Pick the **odd** one for antibiotics

(1) First antibiotic was discovered by chance

(2) Fleming, Chain and Florey established the full potential of penicillin

(3) Use of an antibiotic, extensively saved the wounded soldiers in world war-II.

(4) Antibiotics are the chemicals that are purified from microbes.

101. The most suitable biofertilizer for rice fields among the following can be

(1) *Rhizobium*

(2) *Bacillus*

(3) *Anabaena*

(4) *Azospirillum*

102. All of the following can be achieved by using bacteria **except,**

(1) Large holes in swiss cheese

(2) Ripening of roquefort cheese

(3) Formation of curd

(4) Making dough for idli and dosa

103. Number of nucleotides in the bacteriophage lambda is

(1) 48502

(2) 5386

(3) 97004

(4) 4.6×10^6

104. Choose the **correct** statement from the following.

(1) Adenine pairs with Cytosine through two H-bonds

(2) Thymine pairs with Adenine through two H-bonds

(3) Guanine pairs with Adenine through three H-bonds

(4) Cytosine does not pair with Guanine

105. If the length of human DNA molecule would have been 3.4 metres, the approximate number of base pairs would be

(1) 3.3×10^9

(2) 6.6×10^6

(3) 1.0×10^{10}

(4) 6.6×10^9

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106. Consider the following statements and select the **correct** option.

Statement (A) : A molecule that can act as a genetic material must provide the scope for rapid changes that are required for evolution.

Statement (B) : The association of histone H1 with a nucleosome, leads to the formation of chromatin fibre.

(1) Only statement (A) is true

(2) Only statement (B) is true

(3) Both the given statements are true

(4) Both the given statements are false

107. Who gave the term 'Nuclein'?

(1) Meischer

(2) Crick

(3) Morgan

(4) Griffith

108. In a DNA molecule,

(a) Sugars and phosphates form the backbone.

(b) Total amount of purines and pyrimidines is always equal.

(c) Two chains are coiled in a right-handed fashion.

(d) Fragments are formed when the polymerisation takes place of the template with the polarity 5' → 3' .

(e) The distance between two strands is almost uniform

The **correct** one(s) is/are

(1) (a), (c) and (e) only

(2) (b) and (c) only

(3) (a), (c), (d) and (e) only

(4) (b) and (d) only

109. First phase of translation involves

(1) Aminoacylation of tRNA

(2) Recognition of DNA molecule

(3) Binding of mRNA to ribosome

(4) Recognition of an anticodon

110. Select the **correct** match from the following.

(1)	Hershey and Chase	–	Determined the biochemical nature of the first genetic material
(2)	Meselson and Stahl	–	Gave experimental proof of semiconservative nature of DNA replication
(3)	Maurice Wilkins and Rosalind Franklin	–	Proposed double helix model of DNA
(4)	Avery, Macleod and McCarty	–	Considered genetic material to be a protein

(1) (1)

(2) (2)

(3) (3)

(4) (4)

111. DNA is different from RNA because,

(1) Nucleotide of DNA has nitrogenous base attached at its 1'-position whereas, in case of RNA, it is attached at its 5'-position.

(2) DNA contains Uracil whereas, RNA contains Thymine.

(3) DNA comprises of five-carbon sugars whereas, RNA comprises of six-carbon sugars.

(4) DNA nucleotide lacks a hydroxyl group at the 2'-position whereas, RNA nucleotide has hydroxyl group at the 2'-position

112. A bacteriophage containing radioactive sulphur is allowed to infect *E.coli* that is growing in radioactive phosphorus containing medium. The viruses produced inside this infected bacteria will have

(1) Radioactive protein capsule and non- radioactive genetic material

(2) Radioactive protein capsule and radioactive genetic material

(3) Non-radioactive protein capsule and radioactive genetic material

(4) Non-radioactive protein capsule and non- radioactive genetic material

113. State **True/False (T/F)** for the following statements and select the **correct** option.

(a) The anticodon of a particular tRNA molecule is complementary to the corresponding triplet in rRNA

(b) There are total three types of RNA polymerases in a eukaryotic cell

(c) The order and sequence of amino acids on a polypeptide is defined by the sequence of bases in the mRNA.

(d) 23S rRNA catalyses the formation of peptide bond in bacteria

	(a)	(b)	(c)	(d)
(1)	F	F	T	T
(2)	T	T	F	T
(3)	F	F	T	F
(4)	T	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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114. Match the **column I** with **column II** and select the **correct** option.

	Column I		Column II
a.	mRNA	(i)	Transcribed by RNA polymerase III
b.	tRNA	(ii)	Carries codon sequence, complementary to cistron of template strand
c.	rRNA	(iii)	Plays structural and catalytic role during RNA formation
d.	hnRNA	(iv)	Undergoes the addition of adenylate residues at its 3'-end

(1) a(i), b(ii), c(iv), d(iii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(i), b(iii), c(ii), d(iv)

115. Evaluate the following statements regarding the genetic code.
(a) A cell-free system for protein synthesis helped in deciphering the genetic code.
(b) Some codons code for more than one amino acid, hence the code is degenerate.
(c) The codon UAA, UAG and UGU do not code for any amino acid.
(d) Frameshift mutations demonstrate the triplet and commaless nature of the genetic code.
How many of the above statements is/are **correct** w.r.t. the genetic code?

(1) Four

(2) Two

(3) Three

(4) One

116. During the packaging of DNA helix in eukaryotes,

(1) DNA is held with some proteins in a region called nucleoid.

(2) DNA is organised in large loops, held by acidic proteins.

(3) The negatively charged DNA is wrapped around H1 histone to form a nucleosome.

(4) Beads on a string structure in chromatin is further packaged to form chromatin fibres

117. Which of the following is an **incorrect** match?

(1)	Condensation of chromatin	–	Metaphase stage
(2)	Catalytic RNA	–	28 S rRNA
(3)	ACU	–	Anticodon of stop codon found in t-RNA
(4)	Methyl guanosine triphosphate	–	Unusual nucleotide involved in capping

(1) (1)

(2) (2)

(3) (3)

(4) (4)

118. Identify the **correct** option from the following.

(1) Split – gene arrangement is a characteristic of bacteria.

(2) DNA – dependent RNA polymerase-I is the only enzyme that has the capability to catalyse , elongation and termination of transcription in eukaryotes.

(3) After a mutation at a locus on chromosome, the character of an organism changes due to change in the pattern of protein synthesis.

(4) In prokaryotes, control of the rate of transcriptional initiation is the predominant site for the control of gene expression.

119. Select the **correct** features for human genome, as per the findings of HGP from the following.
(a) It contains 3174.6 million base pairs.
(b) Their is very slight variation in size from gene to gene
(c) Functions are known for over 50 per cent of all the genes.
(d) Repeated sequences make up very large portion of human genome.
(e) Snips occur at about 1.4 million locations.
(f) Repetitive sequences show direct coding functions and shed light on the structure of a chromosome.
The **correct** ones are

(1) (a), (c), (d) and (f)

(2) (d) and (e) only

(3) (a), (b), (c) and (f)

(4) (c), (d) and (e)

120. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A) : In *E. coli*, if a piece of DNA is needed to be propagated during recombinant DNA procedures, it requires a vector.
Reason (R) : There is a definite region in *E. coli* DNA, from where the replication originates and vector provides the additional origin of replication.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

121. Identify the **correct** events that occur during the synthesis of functional mRNA in eukaryotes through transcription and arrange them in the sequential order
(a) Sigma factor recognises its binding site.
(b) Association of termination factor with the template strand.
(c) Uncoiling of DNA helix occurs.
(d) RNA polymerase I gets associated with the initiation factor.
(e) The process of splicing takes place.
(f) Addition of adenylate residues occurs in a template independent manner.
Choose the **correct** answer:

(1) (d) → (c) → (a) → (b) → (e) → (f)

(2) (c) → (a) → (d) → (b) → (f) → (e)

(3) (d) → (c) → (b) → (f) → (e)

(4) (c) → (a) → (d) → (e) → (f)

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122. Which of the following expansions of the abbreviations is **not** correct?

(1) BAC – Bacterial Artificial Chromosome

(2) VNTR – Variable Number of Tandem Repeats

(3) ESTs – Expressed Standard Tags

(4) PCR – Polymerase Chain Reaction

123. Read the statements (A) and (B) and choose the **correct** option.
(A) : In general, carnivores get more adversely affected by competition than herbivores.
(B) : Competition is best defined as a process in which the intrinsic rate of natural increase of one species is significantly lower in the presence of another species.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

124. Match the **List I** with **List II** and select the **correct** option.

	List I		List II
a.	Herbivores-green plants	(i)	Mutualism
b.	Mycorrhizae-roots of higher plants	(ii)	Predation
c.	Epiphytes-on other plants	(iii)	Parasitism
d.	*Cuscuta*-other plants	(iv)	Commensalism

(1) a(i), b(iv), c(iii), d(ii)

(2) a(i), b(iv), c(ii), d(iii)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iii), b(ii), c(i), d(iv)

125. Predation is a type of population interaction which
(a) Is the nature's way of transferring energy across the trophic levels.
(b) Keeps prey population under control.
(c) Forms the basis of the biological control methods, adopted in agriculture.
The **correct** one(s) is/are

(1) (a) only

(2) (b) only

(3) (a) and (b) only

(4) (a), (b) and (c)

126. A population growth model expressed by the equation,
$N_t = N_0 e^{rt}$

(1) Is the more realistic one in nature.

(2) Cannot be exemplified by the population growth of large mammals in the absence of checks.

(3) Supported by Darwin, while he was developing his theory of natural selection.

(4) Represents the unimpeded growth of a population, when the resources are very limited.

127. According to Darwin, potent force in organic evolution was

(1) Mutualism

(2) Interspecific competition

(3) Amensalism

(4) Parasitism

128. Gause's principle of competitive exclusion states that

(1) Competition for the same resources excludes species having different food preferences.

(2) Larger organisms exclude the smaller ones through competition.

(3) Two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

(4) More abundant species exclude the less abundant ones through competition.

129. Read the following statements.
(a) Natality refers to the number of births during a given period in the population that is added to the final density.
(b) Methods of measuring population density depend on the type of organism.
(c) For the purpose of ecological studies, only the organisms that give rise to a population by interbreeding, are considered.
(d) Due to predation pressure, size of a population is not a static parameter.
Which of the above statements are **incorrect**?

(1) (a) and (c)

(2) (b) and (d) only

(3) (a) and (d)

(4) (b), (c) and (d)

130. Pick the **odd** one for parasitism.

(1) The life cycle of endoparasites are more complex due to their extreme specialisation.

(2) Copepods mostly show endoparasitism by infesting the guts of marine fish.

(3) Majority of parasites harm their hosts and reduce the survival capacity of the hosts.

(4) Morphological and anatomical features of endoparasites are greatly simplified

131. Which of the following statements is **correct**?

(1) Abingdon tortoise in Galapagos Islands became extinct due to the lack of shelter on the island.

(2) If a predator is too efficient, it tends to overexploit its prey and maintains its population at a constant level

(3) Connell's elegant field experiments showed the growth of the barnacle *Balanus*, when the barnacle *Chathamalus* was removed from the area.

(4) Resource partitioning mechanism, explained by MacArthur, supports the existence of different species, competing for the same resource in the same area.

132. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A) : Barnacles growing on the back of a whale get benefit in the form of shelter, while whale remains unaffected.
Reason (R) : The type of interaction shown by barnacles and whale is commensalism

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is false but (R) is true

(4) (A) is true but (R) is false

133. Which of the following is **not** true for a mutualistic relationship?

(1) Female wasp using fig fruit as an oviposition site.

(2) Fungal filaments spreading among the roots of higher plants.

(3) Bee performing pseudocopulation with the mediterranean orchid, *Ophrys*.

(4) Blue-green algae interacting with fungus.

134. Identify the **correct** option w.r.t. an urn-shaped population pyramid.

(1) It reflects the status of a population in which stability has been achieved.

(2) It can represent the population of a country in which the individuals in the post-reproductive age group are more than the ones in the reproductive age group.

(3) Such a population keeps on increasing with time, by realising its complete innate potential.

(4) Such population becomes extinct with time, if such growth pattern continues.

135. State **True/False (T/F)** for the following statements and choose the **correct** option.
(a) Absolute population densities are must for ecological investigations, despite the type of sample size and organism.
(b) Visiting flamingoes and zooplanktons in the South American lakes, show competition.
(c) Oysters produce a large number of small- sized offsprings during their lifetime.

	(a)	(b)	(c)
(1)	T	F	T
(2)	F	F	T
(3)	F	T	T
(4)	T	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

ZOOLOGY

136. "Health, for a long time, was considered as a state of body and mind where there was a balance of certain humors". This hypothesis was **not** asserted by

(1) Hippocrates

(2) Early Greeks

(3) W. Harvey

(4) Indian Ayurveda system

137. The School of Cytogenetics and Radiation Research was established at

(1) EFB

(2) CDRI

(3) IARI

(4) RCH

138. Pathogenicity of an organism is
a. Its inability to produce symptoms of a disease
b. Its potential to cause a disease
c. The severity of the disease caused by only a specific strain of that organism
Choose the correct option.

(1) b only

(2) b and c only

(3) a and c only

(4) a, b and c

139. **Assertion (A)** : In humans suffering from elephantiasis, the filarial worms live in lymphatic vessels of lower limbs for many years and causes chronic inflammation in this part.

Reason (R) : Overtime, primarily blood and lymph accumulate between the tissues of the lower limbs which leads to severe fluid retention and causes the underlying tissue to harden and enlarge to extreme proportions.

In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) (R) is true, (A) is false

140. A person initially had fever with chills but he was not treated due to some reasons and later he developed symptoms like gray to bluish colouration of lips and finger nails. Later it was also found that he was suffering from a bacterial disease and his lungs were severely affected due to that.

The causative agent of this infection is

(1) *H. influenzae*

(2) *S.typhimurium*

(3) *P. falciparum*

(4) *S. typhi*

141. Cell-mediated immunity is

(a) A part of the adaptive immune system that uses specialized WBCs

(b) Pathogen specific

(c) Responsible for rejecting foreign tissue transplants

Choose the **correct** option.

(1) (b) only

(2) (a) and (b) only

(3) (a), (b) and (c)

(4) (c) only

142. Select the **incorrect** option w.r.t. interferons.

(1) Protein molecules that protect non-infected cells

(2) They are the part of our innate immune defence system

(3) Their different forms are used in the treatment of different medical conditions such as cancer

(4) They are always secreted by cells infected by pathogenic bacteria

143. Consider the following statements w.r.t. antibodies and select the **incorrect** one.

(1) They belong to a group of lipids called globulins and for this reason, they are also known as immunoglobulins.

(2) Their monomeric forms contain four polypeptide chains, 2 heavy and 2 light chains , hence they are represented as H_2L_2.

(3) A disulfide bond holds each light chain to a heavy chain

(4) Two interchain disulfide bonds are present between the two heavy chains of monomeric form.

144. Match column I with column II.

	Column I		Column II
a.	Naturally acquired active immunity	(i)	Intravenous injection of antibodies
b.	Naturally acquired passive immunity	(ii)	Attenuated pathogens introduced during vaccination, leading to the production of memory cells
c.	Artificially acquired active immunity	(iii)	Transport of IgA antibodies from mother to baby during breast feeding
d.	Artificially acquired passive immunity	(iv)	Exposure to pathogens causing flu that leads to the formation of antibodies

Choose the **correct** option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(ii), c(iv), d(i)

145. Choose the **incorrect** statement.

(1) Tonsils provide the site for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

(2) Lymph nodes act as the site where immature lymphocytes differentiate into antigen-sensitive lymphocytes.

(3) Spleen acts as a filter of blood , trapping foreign substances and activating immune cells to fight against infection.

(4) Some of the secondary lymphoid tissues are located in the mucous membranes of GIT and respiratory tract.

146. Select the option representing the diseases that are characterised by the underactivity of the immune system.

(1) Diabetes mellitus, Rheumatoid arthritis

(2) Myasthenia gravis, Asthma

(3) AIDS, ADA deficiency

(4) Graves' disease, Allergy

147. Identify the protozoal disease that requires a mechanical carrier to transmit the pathogen.

(1) Ascariasis

(2) Malaria

(3) Amoebiasis

(4) Ringworm

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148. Select the correct option to complete the analogy w.r.t. the causative agents of cancer.
Biological agent : Oncogenic viruses :: Chemical agent : _____

(1) X-ray

(2) UV-ray

(3) γ-ray

(4) Nicotine

149. Consider the following statements and select the **incorrect** one.

(1) Mucus membrane acts as the physical barrier and inhibits the entry of microbes.

(2) Natural killer cells are the type of lymphocytes and are the part of the cellular barriers

(3) Our tears contain lysozymes and act as the physiological barrier of innate immunity.

(4) The ability of our body to defend itself in a non-specific manner against bacteria, toxins, viruses and foreign tissue is called adaptive immunity.

150. Choose the **incorrect** match from the following.

(1)	*Atropa belladona*	–	Hallucinogenic properties
(2)	*Cannabis sativa*	–	Affects cardiovascular system
(3)	*Erythroxylum coca*	–	Interferes with the transport of neurotransmitter, dopamine
(4)	*Papaver somniferum*	–	Produces strong stimulants which can treat insomnia

(1) (1)

(2) (2)

(3) (3)

(4) (4)

151. At present, HIV infection cannot be cured completely but, some drugs have been proven to be successful in extending the life of many HIV infected people.
Which of the below mentioned drug therapies can be used for the treatment of a HIV-positive patient?
(a) Reverse transcriptase inhibitor
(b) Anti-retroviral drug therapy
(c) Administration of proteinaceous corticoids
Select the correct option.

(1) (a) only

(2) (a) and (c)

(3) (c) only

(4) (a) and (b)

152. How many of the following are associated with malignant tumors?
a. Starve neighbouring cells
b. Exhibit metastasis
c. Show increased rate of mitosis
d. Immortal
e. Show neoplasticity
Choose the **correct** option.

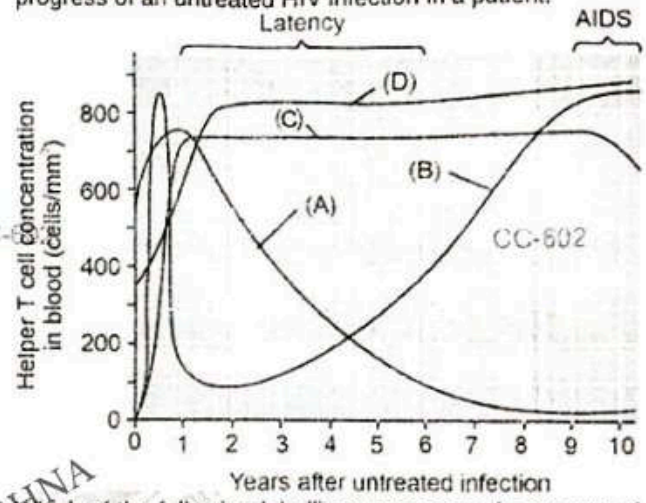
(1) Five

(2) Two

(3) Four

(4) Three

153. Examine the graph given below that illustrates the progress of an untreated HIV infection in a patient.



Which of the following labellings represents the amount of helper T-cells in the blood of this patient?

(1) (A)

(2) (B)

(3) (C)

(4) (D)

154. Within few months of using an antibiotic (named 'X') to treat a strain of *Staphylococcus* infection in a community, it was observed that some new infections were still occuring in the community by the antibiotic 'X' resistant strain of the same bacteria.
How can this phenomenon be best explained?

(1) New strains of bacteria must have evolved *via* large and non-directional variations that showed resistance against 'X'

(2) In response to the antibiotic 'X', *Staphylococcus* got mutated instantly and began making drug-resistant version of proteins for its survival.

(3) Some drug-resistant strains of bacteria must have evolved primarily through random, pre-existing genetic mutations and selection pressure would have allowed them to survive.

(4) All new infections of this antibiotic-resistant strain must have occured due to multi-step large mutations induced in *Staphylococcus*.

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155. DNA sequences in many human genes are very similar to the sequence of corresponding genes present in chimpanzees. This information provides

(1) Morphological evidence, as convergent evolution leads to similarities in the DNA sequences

(2) Palaentological evidence, as chimpanzees evolved from monkeys which are very similar to humans

(3) Embryological evidence, as humans evolved directly from chimpanzees

(4) Biochemical evidence, as humans and chimpanzees share a relatively recent common ancestor

156. When we compare forelimbs of whales and bats, they indicate

(1) Different anatomical structure

(2) Analogy

(3) Common ancestry

(4) Convergent evolution

157. A population is in Hardy-Weinberg equilibrium for a gene with two alleles. If gene frequency of the dominant allele 'A' is 0.6, then, the genotype frequency of the carriers would be

(1) 0.24

(2) 0.42

(3) 0.36

(4) 0.48

158. How many statements given below is/are true w.r.t. natural selection?

(a) Overtime, natural selection can increase the frequency of adaptations which are favourable in a given environment.

(b) Natural selection is a process in which individuals who have certain favourable heritable traits, survive and reproduce at a higher rate than the other members present in the population.

(c) In the event of changes in the environment, natural selection may result in adaptation to new conditions, but cannot give rise to new species in future.

(d) Although natural selection occurs through interactions between individual organisms and their environment, a population does not evolve over time.

Choose the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

159. Arrange the following clades of plants in the correct sequence from earliest to the most recent w.r.t. their origin.

a. Psilophyton
b. Seed ferns
c. Tracheophyte ancestors
d. Dicotyledons
e. Monocotyledons

Select the **correct** option.

(1) a → b → c → d → e

(2) c → a → b → d → e

(3) a → c → d → b → e

(4) d → a → c → e → b

160. Which of the following animals are extinct and considered as the direct ancestors of mammals?

(1) Pelycosaurs

(2) Thecodonts

(3) Sauropsids

(4) Therapsids

161. Selection of the mean character value from three phenotypes of a character is an example of

(1) Stabilising selection

(2) Disruptive selection

(3) Cyclic selection

(4) Directional selection

162. Origin of birds occured _____.

Choose the correct option to fill in the blank.

(1) Before the Silurian period

(2) In the Palaeozoic era

(3) Before the Carboniferous period

(4) After the Triassic period

163. Read the following statements (A – E).

A. The first non-cellular forms of life could have appeared 3 bya.
B. Primitive atmosphere was not reducing.
C. Human evolution represents adaptive radiation.
D. Earth was supposed to have been formed about 4.5 mya.
E. Analogous organs are the result of convergent evolution in separate lineages.

How many of the above statements are true?

(1) Five

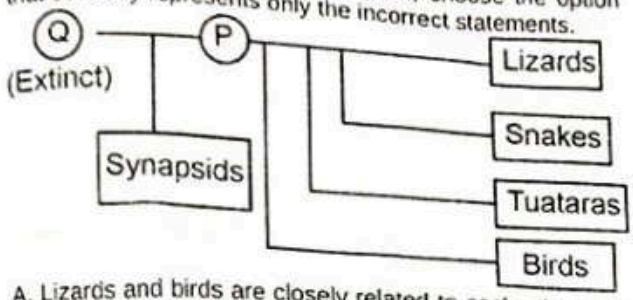
(2) Four

(3) Three

(4) Two

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164. With the help of the given cladogram, choose the option that correctly represents only the incorrect statements.



A. Lizards and birds are closely related to each other than snakes.
B. Tuataras are more closely related to birds than to lizards.
C. Modern day reptiles like snakes and lizards are predecessors of Sauropsids, which are represented by 'Q'.
D. 'Q' represents extinct reptiles that ultimately gave rise to tuataras and birds.

(1) A and D

(2) Only B and C

(3) Only C and D

(4) A, B and C

165. A small wild animal population has a stable size of 10 animals. Frequency of dominant and recessive alleles 'p' and 'q', respectively, are given for 3 generations in the table below.

	Generation-1	Generation-2	Generation-3
Frequency of 'p'	0.5	0.7	0.99
Frequency of 'q'	0.5	0.3	0.01

Choose the option which represents the **correct** inference of the above given case.

(1) Such changes in gene frequencies are never coincidental, rather selection of one allele over other is the result of adaptive radiation.

(2) Chance events like natural calamities can cause allele frequencies to fluctuate unpredictably from one generation to the next

(3) The frequency of dominant alleles always increases due to mutation in every generation, as evolution is a directed process in the sense of determinism.

(4) The continuous change in the gene frequency is the result of gene-flow that occurs to maintain the gene pool of a population at a constant value.

166. **Assertion (A) :** Habitat fragmentation and genetic drift may accentuate variations, leading to appearance of new species and hence evolution.
Reason (R) : Habitat fragmentation splits population into isolated groups while genetic drift randomly changes gene frequencies, ultimately both leading to speciation.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) does not explain (A) correctly

(2) Both (A) and (R) are true, (R) explains (A) correctly

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

167. Which of the following does not explain the process of evolution w.r.t. Darwinsm?

(1) Occurrence of germline variations

(2) Adaptive radiation

(3) Differential reproductive success

(4) Inheritance of acquired characters

168. Consider the statements given below:
Statement A : Few fossils of man-like bones have been discovered in Ethiopia and Tanzania which revealed hominid features, leading to the belief that about 3-4 mya, man-like primates walked in eastern America who did not have an erect posture.
Statement B : Homology is accounted for the idea of branching descent.
Choose the correct option.

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

169. Placental mammals in Australia have variable forms in different habitats which exhibit _____ and these mammals appear to be similar to the corresponding marsupials which can be called as _____.
Choose the option that correctly fills the respective blanks.

(1) Anthropogenic selection, artificial selection

(2) Adaptive radiation, homology

(3) Adaptive radiation, convergent evolution

(4) Analogy, anthropogenic selection

170. Which of the following is **not** correctly matched?

(1)	Greek thinkers	–	Theory of Panspermia
(2)	Louis Pasteur	–	Believed in the theory of spontaneous generation
(3)	Mendel	–	Talked about 'inheritable' factors
(4)	Ernst Heckel	–	Embryological support for evolution

(1) (1)

(2) (2)

(3) (3)

(4) (4)

171. Choose the statement which is **incorrect** w.r.t. amniocentesis.

(1) It is used primarily to detect genetic disorders, but it also provides information concerning the survivability of the foetus.

(2) This procedure involves withdrawing some of the fluid present in the uterine cavity of the mother and analysing maternal cells.

(3) It is used to test the presence of certain sex-linked disorders such as haemophilia.

(4) This test can also reveal the baby's gender

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172. Choose the **correct** match.

(1)	Lactational amenorrhea	–	Decreased level of plasma gonadotrophins prevent conception in nursing mother
(2)	Coitus interruptus	–	Involves abstaining from sexual activity during the luteal phase of the uterine cycle
(3)	Vasectomy	–	Prevents spermiation in males
(4)	LNG-20	–	Non-medicated IUD that suppress sperm motility and fertilising capacity of sperms

(1) (1)

(2) (2)

(3) (3)

(4) (4)

173. How many complications mentioned in the box given below are the possible ill-effects of using contraceptive devices?

Nausea, abdominal pain, breakthrough bleeding, breast cancer, irregular menstrual bleeding

Select the correct option.

(1) Five

(2) Two

(3) Four

(4) Three

174. A female is regularly taking combined oral contraceptive pills that contain both estrogen and progesterone.
Select the correct statement from the following w.r.t. the level of gonadotropins that will be seen in her blood plasma.

(1) Steroidal oral pills will suppress FSH level in her blood due to negative feedback of synthetic ovarian hormones

(2) Due to increased level of the synthetic version of the pregnancy hormones in her plasma, level of LH increases constantly.

(3) The mid-cycle surge of gonadotrophins will be achieved earlier during her reproductive cycles.

(4) Level of steroidal pituitary hormones will never increase in her blood while she is consuming these pills

175. Artificial insemination

(1) Is a process which is performed in the exact similar manner as the standard *in-vitro* fertilisation is done

(2) Is a fertility treatment that involves introduction of sperms to a test-tube containing ova

(3) Is an invasive method of gamete transfer in which the sperms are transferred only into the cervix

(4) Can be used for the couple when the male partner is infertile

176. Choose the **odd** one w.r.t. venereal diseases.

(1) Trichomoniasis

(2) AIDS

(3) Genital warts

(4) Typhoid

177. Consider the options given below regarding sterilization method of birth control and answer as directed thereafter.
a. In females, it can be performed only after a caesarean delivery, not after natural births.
b. Its complete reversal is possible in males.
c. Generally, it is advised to the couples as a terminal method of contraception.
d. After performing tubectomy, secondary oocyte in females will not change into ootid.
Which of the above statements are correct?

(1) a, b and c only

(2) b and d only

(3) b, c and d only

(4) c and d only

178. Which of the following STIs is completely curable, if detected early and treated properly?

(1) AIDS

(2) Trichomoniasis

(3) Hepatitis-B

(4) Genital herpes

179. Choose the correct option w.r.t. the oral contraceptive pill which was developed at CDRI (Lucknow).

(1) It is 'once a month' pill

(2) It prevents ovulation as well as implantation

(3) It contains estrogen and progesterone in the ratio of 2 : 1

(4) It has the lesser side effects as compared to the steroidal contraceptive pills

180. A couple had been unable to conceive for 5 years even after unprotected sexual cohabitation. On evaluation, it was found that the female partner had blocked fallopian tubes, while the male partner had very low sperm count with poor motility. They wanted to opt for ART for conception and thus multiple mature ova were collected from the woman after hormonal stimulation.
Now, amongst the options given below, select the most suitable ART for this couple.

(1) GIFT

(2) Artificial insemination by a donor

(3) *In-vitro* fertilisation followed by embryo transfer

(4) ZIFT

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